

Journal of the Air & Waste Management Association

ISSN: 1096-2247 (Print) 2162-2906 (Online) Journal homepage: www.tandfonline.com/journals/uawm20

Development of a Sample Equilibration System for the TEOM Continuous PM Monitor

Michael B. Meyer, Harvey Patashnick, Jeffrey L. Ambs & Erich Rupprecht

To cite this article: Michael B. Meyer, Harvey Patashnick, Jeffrey L. Ambs & Erich Rupprecht (2000) Development of a Sample Equilibration System for the TEOM Continuous PM Monitor, Journal of the Air & Waste Management Association, 50:8, 1345-1349, DOI: 10.1080/10473289.2000.10464180

To link to this article: https://doi.org/10.1080/10473289.2000.10464180

Published online: 27 Dec 2011.

Submit your article to this journal

Article views: 799

View related articles

Citing articles: 1 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=uawm20

TECHNICAL PAPER

ISSN 1047-3289 *J. Air & Waste Manage. Assoc.* **50**:1345-1349
Copyright 2000 Air & Waste Management Association

Development of a Sample Equilibration System for the TEOM Continuous PM Monitor

Michael B. Meyer, Harvey Patashnick, Jeffrey L. Ambs, and Erich Rupprecht
Rupprecht & Patashnick Co., Inc., Albany, New York

ABSTRACT

In recent years, scientific discussion has included the influence of thermodynamic conditions (e.g., temperature, relative humidity, and filter face velocity) on PM retention efficiency of filter-based samplers and monitors. Method-associated thermodynamic conditions can, in some instances, dramatically influence the presence of particle-bound water and other light-molecular-weight chemical components such as particulate nitrates and certain organic compounds. The measurement of fine particle mass presents a new challenge for all PM measurement methods, since a relatively greater fraction of the mass is semi-volatile.

The tapered element oscillating microbalance (TEOM) continuous PM monitor is a U.S. Environmental Protection Agency (EPA) PM_{10} equivalent method (EQPM-1090-079). Several hundred of these monitors are deployed throughout the United States. The TEOM monitor has the unique characteristic of providing direct PM mass measurement without the calibration uncertainty inherent in mass surrogate methods. In addition, it provides high-precision, near-real-time continuous data automatically. Much attention has been given to semi-volatile species retention of the TEOM method.

IMPLICATIONS

Thousands of TEOM continuous PM monitors have been deployed throughout the world to provide regulatory and special study measurements of PM in ambient air. Some regions of the world, owing to their topography and pollution sources, are characterized by fine particle pollution that can contain an elevated semi-volatile mass fraction. In these areas, it is desirable to increase collection efficiency for this material by maintaining as low an operating temperature as practical while also removing unwanted particle-bound water. The new SES has been developed to allow conditioning of the PM sample stream to a lower humidity and temperature level than the conventional TEOM settings. It is a significant development because it can be applied easily to existing TEOM monitors.

While using this monitor, it is desirable to maintain as low an operating temperature as practical and to remove unwanted particle-bound water. A new sample equilibration system (SES) has been developed to allow conditioning of the PM sample stream to a lower humidity and temperature level. The SES incorporates a special low-particle-loss Nafion dryer. This paper discusses the configuration and theory of the SES. Performance results include high time-resolved $PM_{2.5}$ data comparison between a 30 °C sample stream TEOM monitor with SES and a standard 50 °C TEOM monitor. In addition, 24-hr integrated data are compared with data collected using an EPA $PM_{2.5}$ Federal Reference Method (FRM)-type sampler. The SES is a significant development because it can be applied easily to existing TEOM monitors.

INTRODUCTION

The tapered element oscillating microbalance (TEOM) continuous PM monitor has been in use worldwide for the automatic, near-real-time measurement of PM in ambient air since 1988. It has been designated as a U.S. Environmental Protection Agency (EPA) PM_{10} federal equivalent method since 1990,[1] and also holds a number of additional worldwide approvals by well-known agencies such as the German EPA.[2] EPA federal equivalency designation testing generally requires the 24-hr integrated data from the monitor to compare favorably with data from an EPA Federal Reference Method (FRM) gravimetric sampler. The EPA equivalency designation program does not address sampling periods other than 24 hr.

The TEOM monitor is well known for exacting near-real-time (minute resolution) particle mass measurement based upon the physical law relating spring-mass behavior.[3] It has the unique characteristic of providing direct PM mass measurement without the calibration uncertainty inherent in mass surrogate methods. In addition, it provides high-precision, near-real-time continuous data automatically. By default, the instrument is usually operated using a 3 L/min sample airstream flow rate and a 50 °C filter and sample airstream conditioning temperature.

The TEOM mass sensor is operated at a constant temperature and flow rate so that the net mass change is strictly due to material retained on the sample filter. The conditioning temperature and sample flow rate may be set to alternate values. EPA has approved a set of alternate operating conditions as modifications to the PM_{10} equivalency designation. For example, the TEOM monitor is often operated at a lower temperature (30 °C) and lower flow rate (1 or 2 L/min) in winter, when there is a significant presence of semi-volatile particle mass components such as particulate NO_3^- or organic compounds from wood smoke.

The measurement of fine particle mass (e.g., $PM_{2.5}$) presents a new challenge for all PM measurement methods because a relatively greater fraction of the mass is semi-volatile. More recently, attention has focused on semi-volatile species retention using the TEOM method. Some studies have presented data and discussion concerning the retention of semi-volatile mass on the sample filter of the TEOM monitor relative to the manual gravimetric method.[4-7] Others have taken advantage of its exacting mass measurement principle to study alternate configurations of the TEOM mass sensor that optimize the retention of semi-volatile compounds.[8,9]

In regions where the presence of light molecular weight semi-volatile particles represents an appreciable fraction of the PM mass, it is desirable to maintain as low an operating temperature as practical while also removing unwanted particle-bound water. The primary goal is to introduce an add-on component for the TEOM monitor that can help meet this condition and thus provide a 24-hr integrated PM mass concentration measurement more comparable to that of the FRM. An additional objective is to provide rapid particle equilibration on the sample filter for more representative high time-resolution measurements.

DESCRIPTION

A new system has been developed to maintain the sample stream flowing into the TEOM monitor at a lower temperature and relative humidity than has been possible in the past. This sample equilibration system (SES) continuously conditions the flows of the TEOM monitor, allowing the operation of the instrument year-round at a temperature closer to that of the gravimetric reference method filter equilibration temperature. The system incorporates a Nafion dryer designed for low particle loss,[10,11] and continuously conditions both the sample and bypass flows of the TEOM monitor. Owing to its configuration, the SES is self-regenerating.

As long as the environment in which the TEOM monitor is housed is maintained at 25 °C or less, the instrument can be operated at a temperature set point of 30 °C, even in hot and humid summertime conditions. While TEOM monitors have previously been operated during cooler months at user-definable temperatures between 30 and 40 °C, such operating parameters were not feasible under summertime conditions with higher temperature and absolute humidity levels. The new system enables filter-based mass measurements by the TEOM monitor under consistent, dry conditions, permitting high temporal resolution even under changing summertime temperatures and relative humidities.

The addition of the SES does not require any hardware modifications to TEOM monitors delivered since January 1996 (Series 1400a revision b), and it can be retrofitted onto earlier instruments. The SES fits between the flow splitter that follows the size-selective inlet and the entry point of the TEOM sensor unit (see Figure 1). The tested configuration added, at most, 50 cm to the length of the sample inlet system and conditioned both the sample and bypass flows that together form a total flow rate of 16.7 L/min (1 m^3/hr). Relative humidity sensors monitored the effectiveness of the continuous conditioning process, allowing the TEOM control unit to collect and store this information electronically. The tested system had a diameter of approximately 10 cm, including

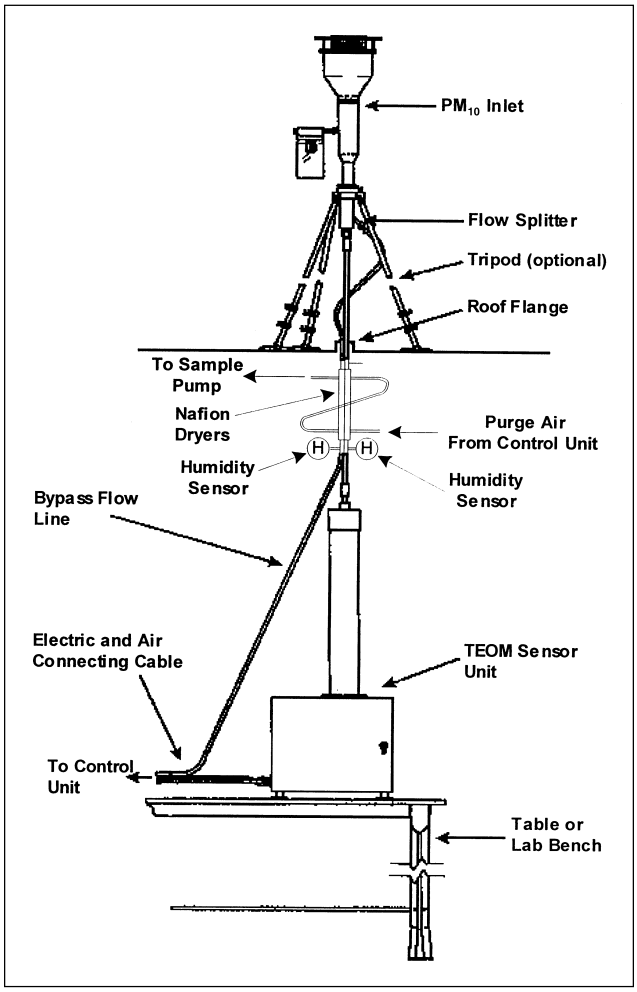


Figure 1. TEOM monitor configuration with SES test system.

both Nafion dryers and the necessary flow interconnections, and did not contain any electromechanical components other than the associated humidity sensors.

Figure 2 shows a schematic of the flow paths in a TEOM system containing the SES. The sample and bypass flows that exit from the flow splitter pass through a pair of Nafion dryers to remove water vapor and reduce humidity levels. From the exit of the SES, the sample flow is routed through the TEOM mass sensor and the bypass flow is routed to the control unit in the usual fashion. The dried sample and bypass flows then proceed through mass flow controllers that maintain a constant volumetric flow rate and are combined just prior to the exit from the TEOM control unit to form a total flow of 16.7 L/min. Due to the pressure drop created at the mass flow controllers, the dried combined air stream forms a lower pressure zone between the controllers and the vacuum pump. This dried, low-pressure merged system flow then becomes the purge flow through the Nafion drier applied to the sample stream, followed by the dryer on the bypass flow. Relative humidity sensors positioned after the dryers provide an ongoing measurement of the effectiveness of the dryers. This information can be collected and logged in the TEOM control unit.

The Nafion dryers incorporated in the tested system were of a specially designed geometry to minimize the potential for particle loss at the entry point. The dryer on the bypass flow line consisted of 50 tubes of 0.025-in. inner diameter (ID), while the sample stream dryer consisted of 18 tubes of 0.06-in. ID. Nafion dryers are similar to shell-in-tube heat exchangers, with an outside shell and strands of Nafion tubing down the center. Nafion material can be used effectively in such applications because of its specific affinity for water. In the presence of a water vapor gradient, Nafion allows the effective passage of moisture. In the SES, the water vapor gradient is created by circulating the dried air that exits from the control unit around the 18 strands of Nafion tubing in the dryer. The Nafion dryers also aid in lowering the sample stream temperature toward that of the air monitoring station or shelter, allowing for an operation setting for the TEOM monitor that can sometimes be lower than the ambient air temperature. The TEOM sample stream temperature setting can be about 5 °C higher than the station temperature.

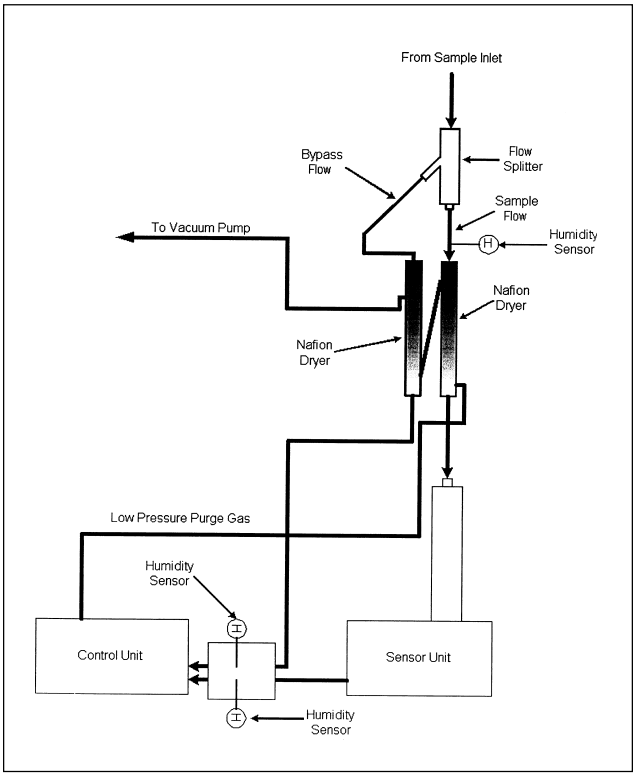


Figure 2. TEOM monitor flow schematic with SES system.

Nafion dryers function on the basis that the sulfonic acid group present in the Nafion tubing has a strong and specific affinity for water. Although water passage through the Nafion is sometimes described as permeation, Nafion dryers do not operate on the same principles as permeation dryers. Nafion is *not* a microporous material that separates compounds on the basis of their molecular size. For example, Nafion dryers can remove water from a hydrogen stream even though the hydrogen molecule is much smaller than water. Pressure is not required to drive the drying process, since the driving force for the reaction is the partial pressure of water vapor.

When water strikes an exposed sulfonic acid group on the surface of the tubing, the surface group initially binds the water. Additional sulfonic acid groups deeper in the wall have less water attached to them and consequently have a higher affinity for water. Water molecules absorbed onto the surface of the tubing are therefore quickly passed on to the underlying sulfonic acid groups until the water reaches the opposite side. This process continues until the water vapor pressure gradient across the tubing wall is eliminated. With a dry purge gas flow over the exterior surface of the Nafion tubing, water vapor is continuously extracted from the gas stream inside the tubing until the sample humidity matches that of the purge gas.

RESULTS

A prototype SES was built and field-tested outside the Rupprecht & Patashnick facility in Albany, NY. The testing period was sufficient to cover a representative range of ambient relative humidity and temperature conditions. One challenging performance evaluation of the SES was possible during a summertime high-temperature, high-relative-humidity episode. Figure 3 shows National Weather Service hourly temperature, dew point temperature, and relative humidity data over a 6-day period, July 30–August 4. The first two days were generally characterized by

daytime temperatures of more than 30 °C and more humid conditions, as indicated by the broader nocturnal relative humidity peaks at or near saturation. An air mass change occurred during August 1, marked by a dew point temperature shift of 8 °C. We can focus on the performance character of the SES by examining the PM mass concentration and exit relative humidity before and after the air mass change.

Figure 4 shows the $PM_{2.5}$ mass concentration from two collocated TEOM monitors, one operated at the default 50 °C conditioning temperature without an SES, and the other set for 30 °C with an SES. Both monitors used a 3 L/min sample air stream flow rate. Also shown is the ambient relative humidity and exit sample air stream relative humidity from the SES. The data indicate that the post-dryer relative humidity stayed reasonably low and stable in the 20–28% range over the entire period. During the higher humidity conditions experienced over the first two days, the SES was able to maintain a sample air stream relative humidity of under 30%. In comparing the temporal patterns in the PM concentration of the two TEOM monitors, it is apparent that the system using the SES generally measured more mass, likely due to better retention of light-molecular-weight semi-volatile material associated with the fine particle mass. In general, use of the dryer component at a reduced conditioning temperature should help reduce the tendency for chemical processes (e.g., chemical dissociation, gas-particle interaction) that can lead to positive or negative biases on sample filters. In addition, the dried sample air stream in the SES system encourages rapid sample equilibration by minimizing longer time-constant equilibrium adjustments of the particle to the surrounding gas stream. This can help provide a better representative short-term mass concentration that includes a measure of semi-volatile components.

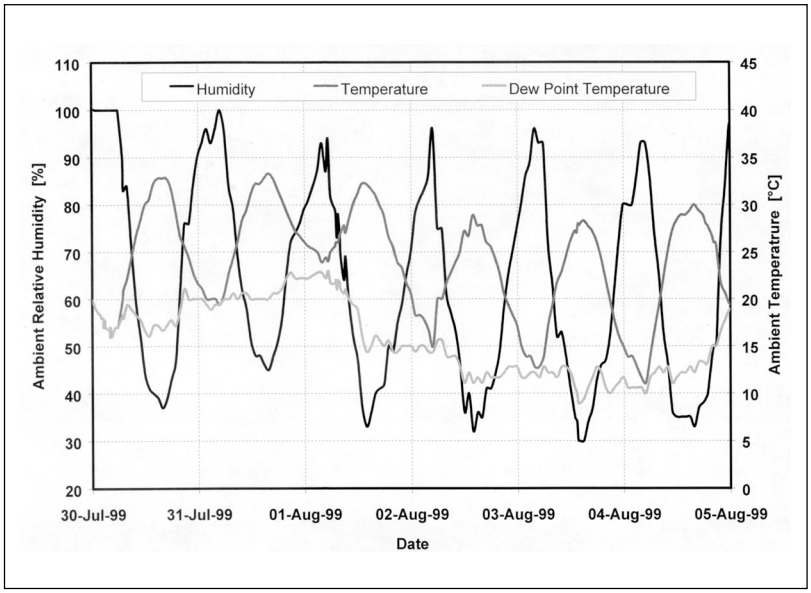


Figure 3. Hourly values of relative humidity, temperature, and dew point temperature (National Weather Service, Albany, NY).

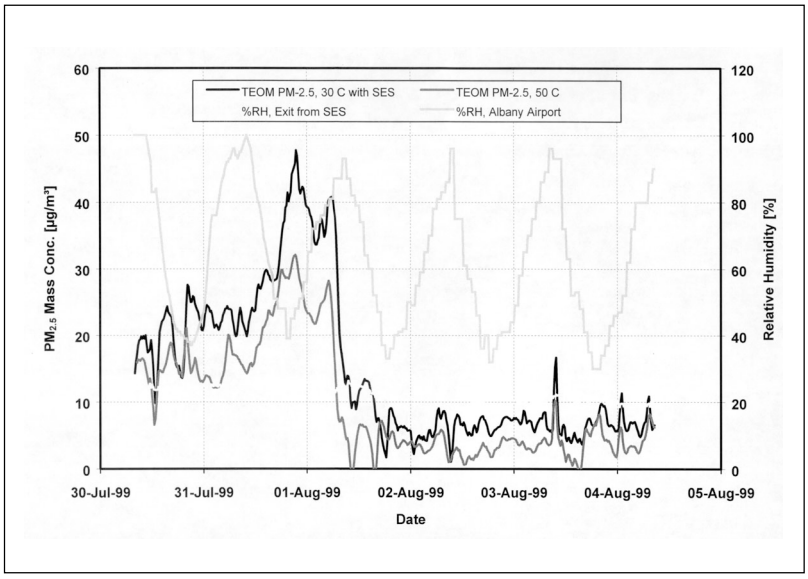


Figure 4. $PM_{2.5}$ mass concentration from two collocated TEOM monitors (30 °C with SES; 50 °C without SES). Ambient relative humidity and TEOM SES exit relative humidity are shown.

A limited comparison of the 24-hr integrated TEOM monitor $PM_{2.5}$ data with collocated EPA FRM-type sampler $PM_{2.5}$ data was made. Figure 5 shows 115 data points collected between June 6 and September 25 from both TEOM monitors (without SES, 50 °C, 3 L/min; with SES, 30 °C, 3 L/min). The two plots show an excellent agreement and correlation between the TEOM monitor with SES and the FRM-type sampler, while at 50 °C and without drying, the data show a difference of about 14% on average. This result suggests that, relative to the FRM-type sampler, a very small percentage of the particle mass sampled during this measurement period in Albany experienced semi-volatile loss. Note that the FRM-type sampler may also exhibit semi-volatile mass loss at the near-ambient sample stream temperature that it maintains. Nonetheless, operating the SES system on a TEOM monitor set at a 30 °C conditioning temperature provides conditioning more conducive to retaining semi-volatile particle mass. However, exact comparison of time-integrated data between automated continuous PM methods and manual gravimetric PM methods can be confounded by the methodologies themselves. Continuous monitors generally utilize a dynamic conditioning approach to provide

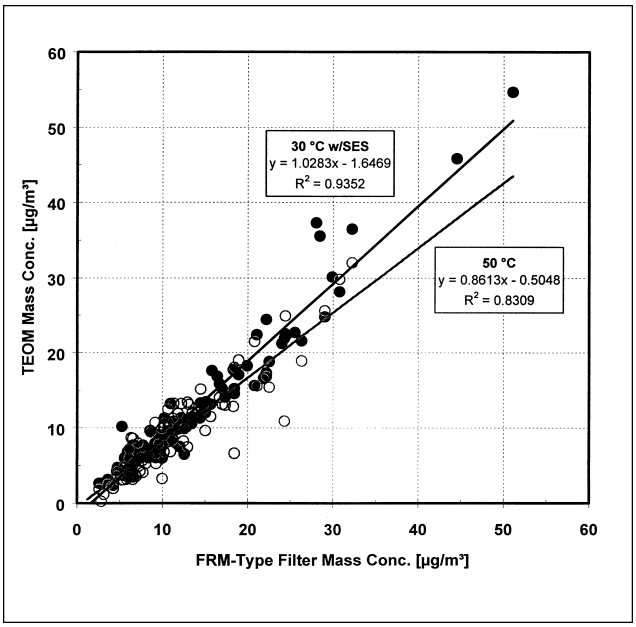


Figure 5. TEOM mass $PM_{2.5}$ mass concentration *vs* FRM-type filter mass concentration. 24-hr averaged values. Filled points from 30 °C TEOM monitor with SES. Open points from 50 °C TEOM monitor.

meaningful near-real-time data. Reference-type gravimetric samplers, however, employ by definition a hybrid (dynamic and static) sample filter conditioning scheme. Reference sample collection often occurs during varying diurnal ambient temperature and relative humidity, while post-sampling filter equilibration is applied in a static environment, first at the varying thermodynamic conditions in the sampler itself, then by the better defined (but different) conditions at the laboratory.

CONCLUSIONS

The SES is most beneficial in regions where a significant fraction of the ambient PM contains light semi-volatile components. It enables more representative ambient PM mass measurements from a TEOM monitor by higher retention of semi-volatile particle mass and reduction in particle-bound water. In addition, the SES can aid in producing more representative high time-resolution data by encouraging rapid particle equilibration on the sample filter.

REFERENCES

1. *Fed. Regist.* **1990**, *55*, 43406.
2. *Gemeinsames Ministerialblatt* **1999**, *22*, 448.
3. Patashnick, H.; Rupprecht, E.G. *J. Air & Waste Manage. Assoc.* **1991**, *41*, 1079-1083.
4. Allen, G.; Sioutas, C.; Koutrakis, P.; Reiss, R.; Lurmann, F.W.; Roberts, P.T. *J. Air & Waste Manage. Assoc.* **1997**, *47*, 682-689.
5. Letters to the Editor, *J. Air & Waste Manage. Assoc.* **1998**, *48*, 195-200.
6. Ayers, G.P.; Keywood, M.D.; Gras, J.L. *Atmos. Environ.* **1999**, *33*, 3717-3721.
7. Mignacca, D.; Stubbs, K. *J. Air & Waste Manage. Assoc.* **1999**, *49*, 1250-1254.
8. Eatough, D.J.; Obeidi, F.; Pang, Y.; Ding, Y.; Eatough, N.L.; Wilson, W.E. *Atmos. Environ.* **1999**, *33*, 2835-2844.
9. Patashnick, H.; Rupprecht, G.; Ambs, J.L.; Meyer, M.B. *Aerosol Sci. & Technol.*, in press.
10. PermaPure, Inc. Product Brochure #100. http://www.permapure.com (accessed April 2000).
11. Dick, W.; Huang, P.; McMurry, P.H. *Characterization of 0.02-µm to 1.0-µm Particle Losses in Perma Pure Dryers: Dependence on Size, Charge, and Relative Humidity*; PTL Publication No. 936; Particle Technology Laboratory, Department of Mechanical Engineering, University of Minnesota: Minneapolis, MN, 1995.

About the Authors

Michael B. Meyer is a senior scientist, Harvey Patashnick is president, Jeffrey L. Ambs is a senior engineer, and Erich Rupprecht is marketing director at Rupprecht & Patashnick Co., Inc. (R&P), 25 Corporate Circle, Albany, NY 12203. Correspondence should be addressed to M.B. Meyer at R&P or sent via e-mail to mbmeyer@rpco.com.